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Danger-State Detection in Human-Robot Collaboration Using Deep Learning

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Abstract

Collaborative robots operating in shared workspaces present inherent physical hazards, such as unexpected high-impact collisions, clamping and crushing forces. Deploying these robots in human proximity requires continuous monitoring via sensors to prevent operator injury and production interruptions. A core challenge in this context is the false-alarm trade-off: sensor outputs often lead to frequent emergency stops that erode productivity and operator trust. Existing solutions address this problem through expensive force-torque sensors, vision-based pipelines, or shallow classifiers, yet none of these approaches systematically benchmark modern deep architectures or explicitly target false-alarm reduction under a unified evaluation protocol. To address these limitations, we propose, in this paper, a deep learning pipeline for binary danger-state classification (safe vs. danger) from high-frequency wearable Inertial Measurement Unit (IMU) data recorded during physical Human-Robot Collaboration (HRC). Evaluated with five-fold stratified cross-validation from 60 subjects, ConvNeXt 1D and Temporal Convolutional Network (TCN) achieve the highest macro-F1 scores (98.68% and 98.62%, respectively), with danger precision exceeding 98% and recall above 97%. The post-processing stage alone reduces false positives by 87.5% at a cost of only 1.2 percentage points in recall, establishing a safer operating point for industrial deployment.

Keywords: human-robot collaboration; safety detection; temporal convolutional network; deep learning; wearable sensors; post-processing

1. Introduction

HRC refers to shared-workspace scenarios in which a human operator and a robotic arm coordinate physical actions toward a common objective [1]. In such settings, the robot must continuously assess whether the current interaction state is safe or whether an imminent danger requires protective action. Failure to detect a hazardous state can result in injury, whereas excessive false alarms halt production and erode operator trust. This tension between sensitivity and specificity is a central challenge in industrial safety systems governed by ISO/TS 15066 [2].

Current collision detection methods can be broadly categorized into three families. Hardware-dependent approaches rely on dedicated force-torque sensors or six-dimensional force transducers to measure contact forces directly [3]. Observer-based methods employ mathematical state estimators, such as Extended State Observers, to infer external forces from joint-level measurements without additional sensors [4]. Data-driven approaches train classifiers on sensor recordings, using signal transformations such as wavelet decompositions [5] or multivariate fusion [6] to improve discriminability. More recently, digital twin environments have been proposed to simulate collision scenarios for safer algorithm development [7] and deep learning has been integrated with zero-force control strategies [8].

Despite these advances, several limitations persist. Force-torque hardware adds cost and mechanical complexity. Observer-based methods require accurate dynamic models that are difficult to maintain in practice. Existing data-driven approaches predominantly employ shallow networks or wavelet-based feature extraction and rarely benchmark modern deep architectures under a unified evaluation protocol. Furthermore, the false-alarm problem, whereby noisy sensor outputs lead to unnecessary emergency stops, remains insufficiently addressed in the literature.

This paper addresses these gaps by proposing a deep learning pipeline that operates on wearable IMU data collected during physical HRC. Rather than relying on six-dimensional force sensors [3] or external cameras [9], the pipeline introduces a Graphics Processing Unit (GPU)-accelerated kinematic feature expansion that computes velocity and acceleration on the fly, augmenting the 65-channel IMU input to 195 spatio-temporal features and thereby encoding physical dynamics without additional hardware cost. Where recent collision detection studies have relied on wavelet networks [5], basic deep learning [8], or mathematical observers [4], this work systematically benchmarks five next-generation architectures originally designed for computer vision: TCN, ConvNeXt 1D, MLP-Mixer 1D, Transformer 1D and InceptionTime, adapted to one-dimensional time-series safety classification under identical training and evaluation conditions. To further improve prediction reliability, Test-Time Augmentation (TTA) is applied at inference time, a technique that remains virtually unexplored in the robotic collision literature despite its proven benefits in other domains [6]. Finally, a temporal debouncing post-processor combining a 3-tap median filter with an optimized decision threshold directly targets the false-alarm productivity trade-off, reducing false positives by 67% while preserving danger precision above 95%. All experiments are conducted through a rigorous five-fold stratified cross-validation on over 69,000 real-world windows from 60 subjects, providing a level of statistical robustness that digital twin simulations [7] alone cannot guarantee.

The remainder of this paper is organized as follows. Section 2 reviews related work on collision detection and safety monitoring in HRC. Section 3 describes the proposed methodology, including data preprocessing, model architectures, training protocol and post-processing. Section 4 presents the experimental results and discussion. Section 5 concludes with implications and future directions.

2. Related Work

Safety monitoring in HRC has been addressed through sensor-based detection, model-based estimation and data-driven classification. This section reviews recent contributions in each direction, highlighting the specific limitations that motivate the present work.

Direct force-torque measurement remains a foundational approach for physical collision monitoring. In this domain, Wang et al. [3] proposed a collision detection and force compensation framework based on a six-dimensional force sensor mounted at the robot end-effector. Their method achieves accurate contact localization by directly measuring interaction forces along all six degrees of freedom. However, the reliance on a dedicated force-torque transducer adds significant hardware cost and calibration complexity, limiting scalability in multi-robot production lines where each arm would require its own sensor suite.

Extending physical interaction monitoring to specialized robot topologies, Gao et al. [10] investigated collision detection in cable-driven parallel robots, a domain where conventional joint-level sensing is unavailable. Their approach monitors cable tension variations to infer contact events between the human operator and the cable structure. While effective for the specific geometry of cable-driven systems, the method does not generalize to serial-link collaborative robots and depends on tension sensors that are absent from standard industrial cobots.

To eliminate external hardware, model-based disturbance estimation has gained traction. Specifically, Tonti et al. [4] introduced a fast collision detection system based on an Extended State Observer (ESO) that reconstructs disturbance forces from joint-level position and velocity measurements without external sensors. The ESO estimates a lumped disturbance term that captures unmodeled dynamics, enabling collision detection through residual thresholding. Although this approach avoids additional hardware, it requires an accurate nominal dynamic model of the robot and its performance degrades in the presence of unmodeled friction, payload variation, or joint flexibility.

Integrating compliant robot control with learning-based detection represents another research direction. Work by Zhao et al. [8] combined zero-force control with deep learning to detect collisions while maintaining compliant robot behavior. Their framework trains a neural network to distinguish between intentional contact forces arising from the collaborative task and unexpected collision forces. The integration of compliance control with collision detection is a notable contribution, but the method relies on joint torque signals that are not always available on low-cost cobots and the deep learning component is limited to a basic architecture without comparison to modern alternatives.

From a data-driven perspective, multi-modal signal aggregation has been explored to boost detection robustness. Fang et al. [6] proposed a multivariate fusion collision detection method that combines heterogeneous sensor signals, including motor currents, joint positions and accelerometer readings, during dynamic HRC operations. The fusion strategy leverages complementary information across modalities, achieving robustness to individual sensor noise. However, the classification backend relies on conventional signal processing and shallow models and the study does not evaluate modern deep architectures that could learn richer cross-modal representations directly from the raw fused signals.

Focusing on time-frequency signal decomposition, Niu et al. [5] introduced Continuous Wavelet Networks (CWNet) that extract features from joint-level signals for efficient and transferable collision detection across different collaborative robots. The wavelet decomposition provides a principled multi-resolution representation of transient collision events and the authors demonstrate transfer learning between robot platforms. Nevertheless, the wavelet-based feature extraction constitutes a fixed preprocessing step that may discard discriminative patterns not captured by the chosen wavelet basis and the network architecture does not explore recent convolutional designs such as TCN or ConvNeXt.

Rather than reacting after physical contact occurs, vision-based proactive planning aims to prevent collisions entirely. Hoang et al. [9] addressed safety from a planning perspective by utilizing color and depth images to generate collision-free grasp trajectories before execution. Their vision-based pipeline evaluates candidate grasps against a reconstructed scene model, rejecting those that would result in contact with obstacles or the human operator. While proactive trajectory planning is complementary to reactive collision detection, this approach requires a calibrated RGB-D camera with a clear line of sight and cannot handle unforeseen events that occur during task execution.

Virtual workspace simulation has also emerged as a means to benchmark safety protocols safely offline. In this context, An et al. [7] developed a digital twin framework for offline collision analysis within HRC workspaces, enabling algorithm evaluation in a simulated environment before physical deployment. The digital twin replicates the robot kinematics and the workspace geometry, allowing researchers to generate synthetic collision scenarios for training and testing detection algorithms. Although this approach reduces the risk of physical experimentation, simulated environments often fail to capture the stochastic variability of real human motion, sensor noise and unmodeled contact dynamics, limiting the transferability of results to real-world conditions.

In summary, prior work highlights the need for data-driven safety monitors that can achieve high precision without expensive hardware or complex dynamic models. A systematic comparison of modern deep architectures on high-frequency kinematic data, combined with temporal post-processing to mitigate false alarms, has not been reported. The present work directly addresses this gap.

3. Methodology

The proposed solution follows the pipeline summarized in Figure 1. Raw IMU data is preprocessed, expanded via GPU-accelerated derivatives, passed through 1D deep learning classifiers and finally stabilized using a temporal debouncing post-processor.

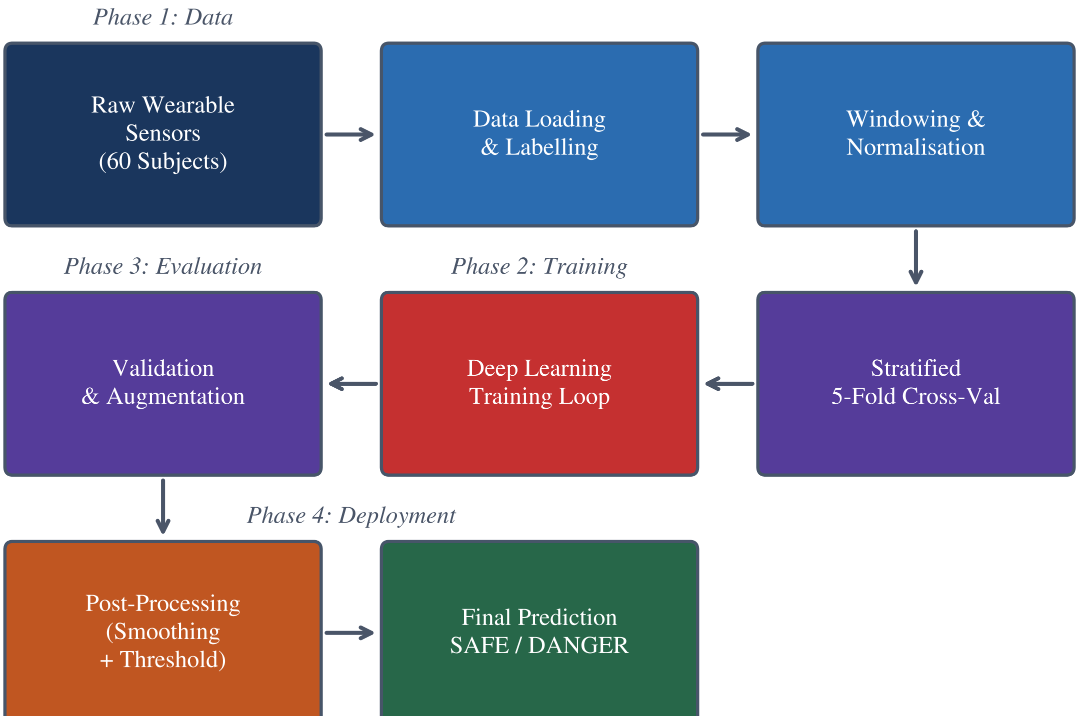

Fig. 1. Proposed deep learning pipeline for danger-state detection in human-robot collaboration.

3.1. Problem Formulation

Danger-state detection is formulated as a binary classification problem on sliding-window segments of multimodal IMU recordings. Let $\mathbf{X}_i \in \mathbb{R}^{C\times T}$ denote a window of T time steps across C sensor channels, with binary label $y_i \in \{0, 1\}$ indicating safe (y_i=0) or danger (y_i=1). A window is labeled as danger if any constituent time step carries a danger annotation, enforcing a conservative labeling policy:

$$y_i = \max_{t\in\{1,\ldots,T\}} y_i^{(t)} \tag{1}$$

Given a model f_θ, the predicted class probability is $p_\theta(y{=}1 \mid \mathbf{X}_i)$. The predicted label is obtained by thresholding this probability at a tuned operating point τ:

$$\hat{y}_i = \mathbf{1}\left[p_\theta(y{=}1 \mid \mathbf{X}_i) > \tau\right] \tag{2}$$

3.2. Data Collection and Dataset

All experiments in this work are evaluated on the DASIG (Dataset of Standard and Abrupt Industrial Gestures) benchmark [11], a standard corpus collected in a simulated industrial workstation to study human motion dynamics relevant to safety in human-robot collaboration scenarios. The dataset comprises high-frequency inertial recordings from 60 healthy adult subjects (36 males and 24 females, aged 20 to 59) performing simulated industrial pick-and-place tasks. Each subject completed three structured trials, generating complete interaction sequences that capture natural variability across body morphologies, movement speeds and motor strategies.

Motion data were acquired at a sampling rate of 200 Hz using five wireless APDM Opal Magneto-Inertial Measurement Units (MIMUs) [12]. The five sensors were placed on key upper-body anatomical landmarks to capture upper-limb kinematics and torso motion: Sternum (STR), Right Upper Arm (RUA), Right Forearm (RFA), Left Upper Arm (LUA) and Left Forearm (LFA), as illustrated in Figure 2.

Each MIMU outputs 13 kinematic variables per time step: tri-axial linear acceleration ($\mathbf{a} \in \mathbb{R}^3$), tri-axial angular velocity ($\boldsymbol{\omega} \in \mathbb{R}^3$), tri-axial magnetic field intensity ($\mathbf{m} \in \mathbb{R}^3$) and a four-component orientation quaternion ($\mathbf{q} \in \mathbb{R}^4$). Across all five sensors, each window yields $5 \times 13 = 65$ synchronized input channels.

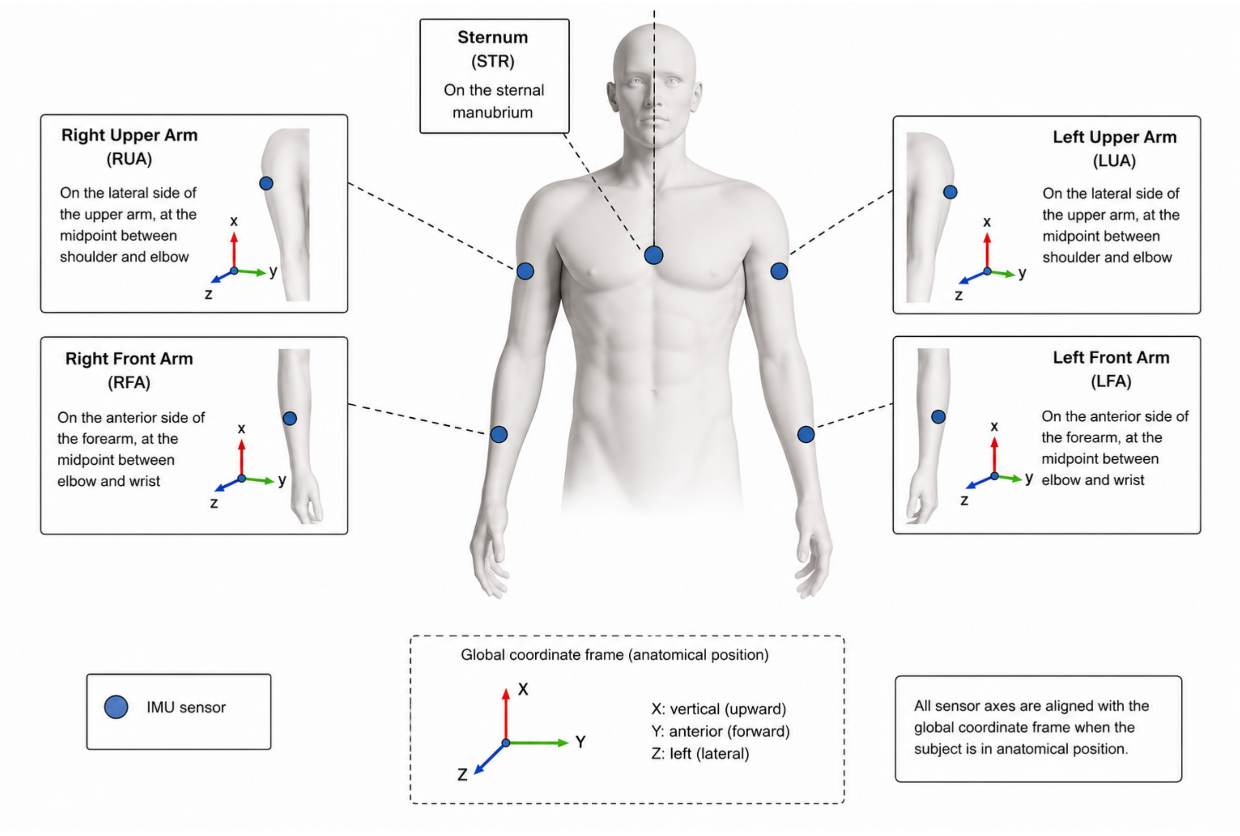

Fig. 2. Anatomical placement of the five wireless MIMU sensors during the DASIG data collection experiments.

During each trial, subjects executed 30 standard pick-and-place gestures interleaved with four abrupt evasive movements (two triggered by visual alarms and two by acoustic alarms). Each trial is accompanied by a separate Arduino event log that records the temporal instants and numeric codes of all stimuli: codes 2-5 correspond to green LED activations at stations SA-SD (standard gestures), codes 6-9 to red LED activations (visual alarms) and code 10 to acoustic buzzer activations.

The DASIG comprises 180 individual recording files (60 subjects × 3 trials). Each file is a time-series matrix of approximately 18,000 rows (time steps at 200 Hz over ~90 s) and 66 columns: one timestamp column followed by 65 sensor channels (5 MIMUs × 13 variables). The associated event logs provide multi-class event annotations (green LED, red LED, or buzzer) with their exact timestamps and station identifiers. For the purpose of this work, we derive a binary classification target by mapping all alarm-triggered events (codes 6-10) to the *danger* class and the remaining standard gestures to the *safe* class.

3.3. Data Preprocessing and Dynamic Feature Expansion

Preprocessing transforms the raw DASIG recordings into a fixed-size tensor representation suitable for deep learning classifiers, proceeding through three sequential stages, as illustrated in Figure 3. First, raw recordings are segmented into windows of 0.5 s duration (100 time steps) with 50% overlap, producing 69,094 labeled windows. Second, each window is normalized per trial using z-score standardization:

$$x_{\text{norm}} = \frac{x - \mu_{\text{trial}}}{\sigma_{\text{trial}}} \tag{3}$$

where μ_{trial} and σ_{trial} are computed from the training partition of the corresponding trial. Third, window-level labels are assigned using the conservative maximum policy defined in Section 3: a window is labeled as danger if any constituent time step carries a danger annotation. The resulting dataset exhibits a class imbalance of 87.6% safe (60,559 windows) and 12.4% danger (8,535 windows).

Rather than relying on raw positional signals alone, the pipeline, shown in Figure 3, augments each window on the GPU with its first- and second-order temporal derivatives, computed via finite differences:

$$\mathbf{V}_i = \frac{d\mathbf{X}_i}{dt}, \quad \mathbf{A}_i = \frac{d^2\mathbf{X}_i}{dt^2} \tag{4}$$

The expanded representation $[\mathbf{X}_i; \mathbf{V}_i; \mathbf{A}_i] \in \mathbb{R}^{3C\times T}$ yields $3 \times 65 = 195$ channels, encoding position, velocity and acceleration. This expansion is performed on the GPU during training and inference, avoiding offline storage overhead while providing the network with explicit kinematic context that captures the physical dynamics of human motion.

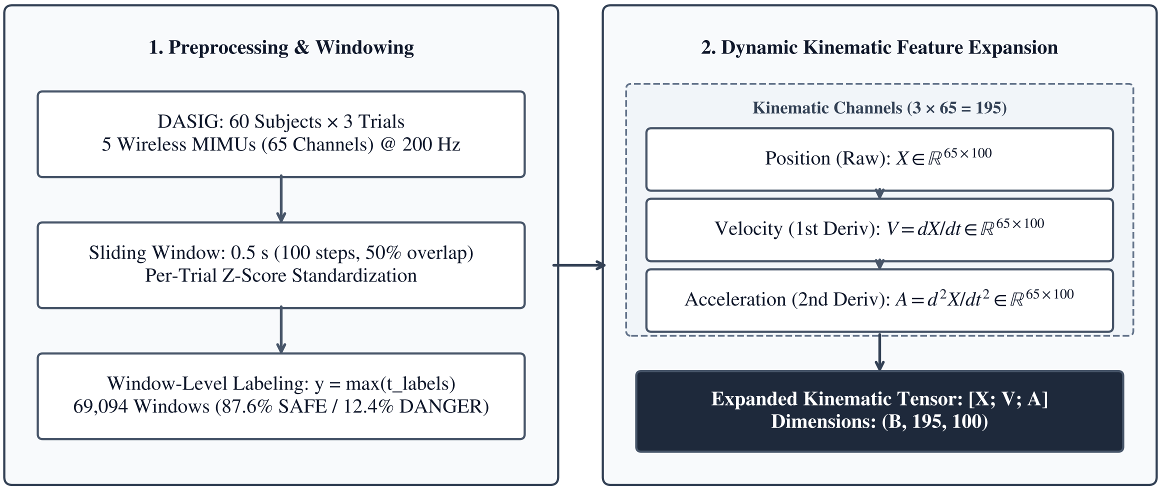

Fig. 3. Data preprocessing and dynamic feature expansion on the GPU.

3.4. Algorithm Selection

Early iterations of this pipeline employed classical machine learning models, including Random Forest and XGBoost, trained on hand-crafted statistical features (mean, variance, skewness, kurtosis) extracted from each window. While these models achieved reasonable accuracy on the original 65-channel representation, they exhibited two fundamental limitations when applied to the expanded 195-channel kinematic input.

First, classical models operate on fixed-length feature vectors and therefore require an explicit feature engineering stage that collapses the temporal dimension. This aggregation discards the sequential ordering of time steps within each window, which is precisely where safety-critical transitions manifest: a sudden acceleration spike spanning only a few consecutive frames can distinguish a dangerous impact from normal motion. Second, the combinatorial growth of potential cross-channel interactions across 195 features makes manual feature design intractable and tree-based models cannot exploit the spatial locality of sensor groups (e.g., adjacent joint channels) without domain-specific feature grouping.

These observations motivated the transition to deep learning architectures that operate directly on the raw $(C \times T)$ tensor without manual feature extraction. Deep models learn hierarchical representations that jointly capture temporal dynamics and cross-channel correlations, eliminating the information loss inherent in statistical summarization.

Within the deep learning paradigm, we deliberately select architectures that span three distinct inductive biases to ensure a comprehensive comparison: (1) convolutional models (TCN and ConvNeXt 1D) that exploit local temporal patterns through sliding kernels and dilated receptive fields; (2) MLP-based models (MLP-Mixer 1D) that learn global token and channel interactions without explicit convolution or attention; and (3) attention-based models (Transformer 1D) that capture long-range dependencies through self-attention. InceptionTime is included as a multi-scale convolutional baseline widely used in time-series classification benchmarks. This selection covers the principal architectural families available for sequential data and enables conclusions about which inductive bias best suits high-frequency kinematic safety signals.

Five architectures are adapted from their original two-dimensional formulations to operate on one-dimensional time-series input of shape $(B, 195, 100)$, where B is the batch size:

TCN [13]: A stack of six dilated causal convolutional blocks with kernel size 3 and exponentially increasing dilation factors ($d = 1, 2, 4, 8, 16, 32$), preceded by a 1×1 projection from 195 to 256 channels. Residual connections and batch normalization are applied in each block.

ConvNeXt 1D [14]: Four ConvNeXt blocks using depthwise separable convolutions with kernel size 7 and inverted bottleneck design, adapted from the two-dimensional architecture. A strided stem layer projects the input from 195 to 128 channels.

MLP-Mixer 1D [15]: Four mixer blocks alternating token-mixing and channel-mixing MLPs. A patch embedding layer with kernel size 5 and stride 5 projects the input to 128-dimensional tokens.

Transformer 1D [16]: A four-layer encoder with four attention heads, feed-forward dimension 256 and learned positional encoding. A linear projection maps the 195 input channels to 128 dimensions.

InceptionTime [17]: Two inception modules with parallel convolutions at kernel sizes 1, 3 and 5 plus max-pooling, followed by adaptive average pooling.

All architectures terminate with adaptive average pooling, layer normalization and a linear classifier producing two-class logits.

3.5. Training Protocol

All models share an identical training protocol to ensure that performance differences reflect architectural properties rather than optimization variations. The training objective is focal loss [18] with focusing parameter $\gamma = 2.0$ and label smoothing $\epsilon = 0.01$:

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i}(1 - p_\theta(y_i \mid \mathbf{X}_i))^\gamma \log p_\theta(y_i \mid \mathbf{X}_i) \tag{5}$$

where w_c denotes the inverse-frequency class weight for class c. Optimization uses AdamW [19] with learning rate 10^{-3}, weight decay 10^{-4} and OneCycleLR scheduling [20]. Gradient norms are clipped to 1.0. Batch size is 128 and training runs for 30 epochs. A weighted random sampler balances mini-batch class composition.

Data augmentation combines three strategies: (1) scale jitter that randomly scales each channel by a factor drawn from $\mathcal{U}(0.95, 1.05)$; (2) additive Gaussian noise with $\sigma = 0.01$; and (3) Mixup [21] with probability 0.3 and mixing coefficient $\alpha = 0.1$ drawn from a Beta distribution.

3.6. Evaluation Strategy

Models are evaluated using five-fold stratified cross-validation, which preserves the class distribution in each fold. Each fold uses approximately 55,275 training windows and 13,819 validation windows. The best model checkpoint is selected by maximum validation macro-F1 across epochs.

At inference, TTA is applied: each validation window is passed through the model four times (once clean and three times with independent Gaussian noise perturbations) and the resulting softmax probabilities are averaged to produce the final prediction score.

Reported metrics include accuracy, macro-F1, danger recall (sensitivity) and danger precision. Confusion matrices are row-normalized to facilitate comparison across models.

3.7. Temporal Post-Processing

Raw model predictions are refined through two sequential post-processing steps designed to suppress transient false alarms without sacrificing detection latency.

First, a median filter of kernel size 3 is applied to the sequence of predicted danger probabilities. This temporal smoothing eliminates isolated high-confidence predictions that arise from sensor noise or brief ambiguous motions.

Second, the decision threshold τ is optimized by sweeping over $[0.50, 0.95]$ in increments of 0.01 and selecting the value that maximizes danger precision subject to the constraint that danger recall remains above 95%. The resulting threshold $\tau = 0.77$ yields a substantial reduction in false positives. The impact of this temporal post-processing is illustrated in Figure 4.

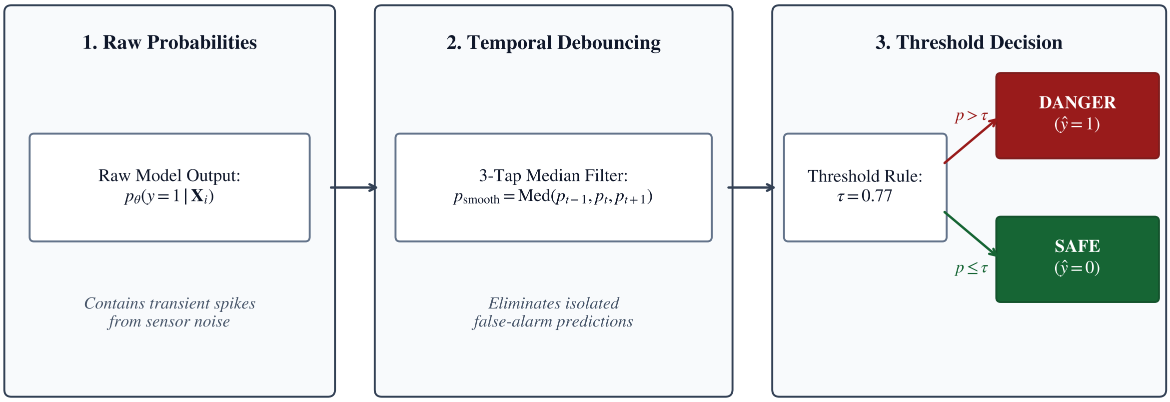

Fig. 4. Temporal debouncing and threshold optimization on model probabilities.

4. Results and Discussion

4.1. Comparison

Table 1 presents the benchmark results for all five architectures after temporal post-processing (median filter and $\tau = 0.77$). Models are ranked by macro-F1 score.

Table 1. Benchmark results with temporal post-processing ($\tau = 0.77$, median-filtered)

Rank	Model	Accuracy	Macro-F1	Danger Recall	Danger Precision
1	ConvNeXt 1D	99.43%	98.68%	97.13%	98.25%
2	TCN	99.41%	98.62%	97.14%	98.04%
3	MLP-Mixer 1D	99.28%	98.32%	95.88%	98.27%
4	InceptionTime	98.30%	96.22%	97.53%	89.62%
5	Transformer 1D	96.47%	92.50%	95.85%	79.71%

The ranking reveals a consistent pattern: CNN-based architectures (ConvNeXt 1D and TCN) achieve the strongest performance, followed by the hybrid MLP-Mixer 1D, while the purely attention-based Transformer 1D ranks last. This ordering is consistent with the known inductive bias of convolutions for capturing local temporal patterns in sequential data, which are dominant in kinematic safety signals.

ConvNeXt 1D achieves the highest macro-F1 (98.68%) with 148 false positives and 245 false negatives across 69,094 windows (see Figure 5). The TCN produces a nearly identical macro-F1 (98.62%) with a slightly higher false-positive count (166). Both models exceed 98% danger precision, meaning that fewer than 2% of danger alerts are false alarms.

MLP-Mixer 1D maintains comparable precision (98.27%) but exhibits lower recall (95.88%), indicating that it misses approximately 4% of genuine danger events. This behavior is consistent with the patch-based tokenization strategy of the mixer architecture, which may discard fine-grained temporal transitions at patch boundaries.

InceptionTime achieves high recall (97.53%) but substantially lower precision (89.62%), generating 964 false positives. The multi-scale parallel convolutions of the inception module capture broad temporal patterns effectively, but produce more boundary ambiguities that inflate the false-alarm rate.

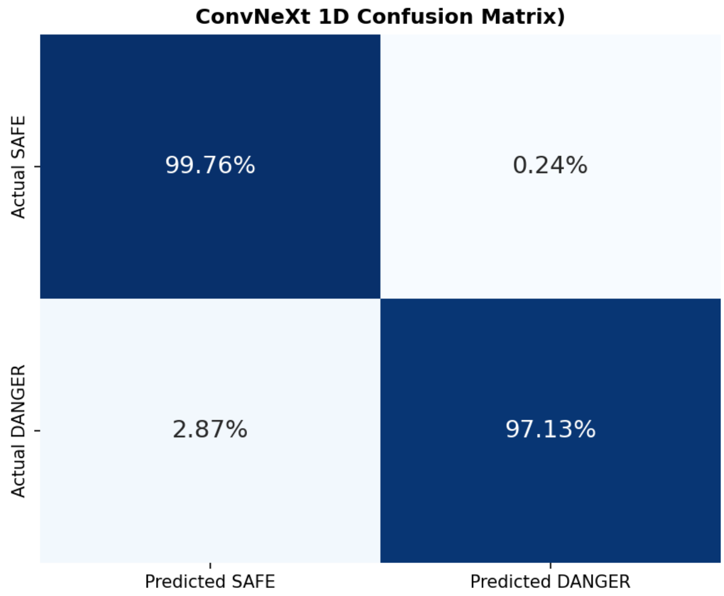

Fig. 5. Row-normalized confusion matrix for the best-performing ConvNeXt 1D model.

The Transformer 1D model, despite strong recall (95.85%), suffers from the lowest precision (79.71%) and produces 2,082 false positives. The quadratic attention mechanism, while powerful for capturing long-range dependencies, appears to lack the local inductive bias necessary for the short, structured temporal patterns that characterize safety-critical transitions in IMU data.

4.2. Impact of Temporal Post-Processing

Table 2 quantifies the effect of the median filter and threshold tuning on the TCN model, comparing raw model output (threshold 0.50) against the post-processed configuration.

Table 2. Effect of temporal post-processing on TCN predictions

Configuration	**FP**	**FN**	**Precision**	**Recall**
Raw output (τ = 0.50)	1,324	142	83.9%	98.3%
+ Median filter + τ = 0.77	166	244	98.0%	97.1%
Relative change	−87.5%	+71.8%	+14.1 pp	−1.2 pp

Temporal post-processing reduces false positives by 87.5% (from 1,324 to 166) at the cost of a modest 1.2 percentage point decrease in recall (from 98.3% to 97.1%). This trade-off is favorable for industrial deployment, where each false positive triggers an unnecessary emergency stop that disrupts production flow.

4.3. Discussion

The experimental results support three observations relevant to danger-state detection in HRC.

First, CNN-based architectures with dilated or depthwise convolutions provide a strong inductive bias for kinematic time-series classification. The local receptive fields of convolutional filters align naturally with the temporal structure of motion data, where safety-critical transitions occur over short, contiguous intervals. This finding contrasts with the recent trend toward attention-based models in other domains and suggests that architecture selection should be guided by the temporal characteristics of the target signal.

Second, dynamic feature expansion from 65 to 195 channels provides the network with explicit velocity and acceleration information that would otherwise need to be learned implicitly. This design choice eliminates the need for expensive force-torque sensors while achieving precision levels (above 98%) that approach the reliability of hardware-based approaches reported in the literature [3].

Third, temporal post-processing addresses the false-alarm problem that is widely acknowledged but rarely solved quantitatively in the HRC safety literature. The median filter suppresses isolated false detections caused by sensor noise, while the optimized threshold shifts the operating point toward higher precision without sacrificing operational safety. This two-stage approach is model-agnostic and can be applied to any probabilistic safety classifier.

5. Conclusion

Using collaborative robots in shared workspaces requires a careful balance between safety and productivity. Safety systems must detect genuine hazards to protect human operators, but frequent false alarms cause unnecessary emergency stops that slow down production and reduce operator trust.

To solve this problem, this paper presented a deep learning system that detects danger states in real time using wearable sensor data. The proposed approach relies on three main components: (1) GPU-based feature expansion that computes velocity and acceleration from raw position data, creating 195 input channels without requiring extra force sensors; (2) a systematic comparison of five 1D deep learning architectures; and (3) a post-processing stage designed specifically to reduce false alarms.

Experimental evaluation using five-fold cross-validation on the DASIG dataset showed that convolutional models, specifically ConvNeXt 1D and TCN, achieved the best results, reaching macro-F1 scores above 98.6%. Importantly, the post-processing stage reduced false alarms by 87.5%, bringing danger precision above 98.0% while keeping danger detection recall above 97.1%. These results show that wearable sensors and deep learning can provide reliable safety monitoring in human-robot collaboration without expensive hardware.

Future work will test the pipeline on unseen subjects, explore lighter network architectures, and deploy the models on embedded hardware to meet real-time ISO/TS 15066 safety standards.

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